



University of sigmas

Oiowe chady

Adam Biel, Mateusz Jurach, Karol Źeleźnik

2024-03-17

1 **Templatki**

2 **Struktury danych**

3 **Optymalizacje**

Templatki (1)

.bashrc 11 lines

```
c() {
    g++ -std=c++20 -Wall -Wno-sign-conversion -Wextra -Wshadow -
        Wconversion -Wfloat-equal \
        -fsanitize=address,undefined -DLOCAL -DDEBUG -D_GLIBCXX_DEBUG
        -ggdb3 $1.cpp -o $1
}

nc() {
    g++ -std=c++20 -DLOCAL -O3 -static $1.cpp -o $1
}
```

```
alias cp='cp -i'
alias mv='mv -i'
```

.vimrc 7 lines

```
set nocompatible
filetype indent on
set autoindent
set number
syntax on
set ts=4 sw=4
set rnu
```

main.cpp 23 lines

```
#include <bits/stdc++.h>
using namespace std;
#ifdef DEBUG
auto&operator<<(auto &o, pair<auto, auto> p) {o << "(" << p.
    first << ", " << p.second << ")"; return o;}
auto operator<<(auto &o, auto x)->decltype(x.end(), o) {o<<"{";
    for(auto e : x) o<<e<<" "; return o<<"}";}
#define debug(X) cerr << "["#X"]": " << X << '\n';
#else
#define cerr if(0)cout
#define debug(X) ;
#endif
using ll = long long;
#define all(v) (v).begin(), (v).end()
#define ssize(x) int(x.size())
#define fi first
#define se second
#define mp make_pair
#define eb emplace_back

int main() {
    ios_base::sync_with_stdio(false); cin.tie(nullptr);

    return 0;
}
```

gen.cpp
Opis: Dodatek do generowania testów, lub innego losowania. Ziarno można zmieniać np. rng.seed(atoi(argv[1]));

```
mt19937 rng(random_device{}());
```

```
1 int rd(int l, int r) {
    return uniform_int_distribution<int>(l, r)(rng);
}
1 }
```

3 **Struktury danych (2)**

wavelet.cpp
Opis: $O(\log^2 \text{ALPH})$ czasowo, $O(n \log \text{ALPH})$ pamięciowo. Constructor przyjmuje ciąg na którym rozpinamy drzewo, rank(l, r, k) zwraca k-tą najmniejszą liczbę na przedziale [l;r]. (**Uwaga!**) pamięć około 262 Mb, dla $N, Q \leq 2 \cdot 10^5$.

```
//funkcja cnt jest w  $O(\log)$  co daje cale operacje w  $O(\log^2)$ ,
//da sie cnt w czasie stalym — do dopisania
struct wavelet_tree {
    const static int ALPH = (1<<18)-1;
    struct Node {
        int low, high, mid;
        vector<int> v;
        Node *left, *right;

        Node() {
            low = 0, high = ALPH;
            left = right = nullptr;
            mid = (low+high)/2;
        }

        Node (const vector<int> &x, const vector<int> &t, int a =
            0, int b = ALPH): low(a), high(b), v(x) {
            mid = ((ll)low+high)/2;
            if(ssize(x) && low != high) {
                vector<int> div[2];
                for(auto e : v) div[t[e]<=mid].eb(e);
                if(ssize(div[1])) left = new Node(div[1], t, low, mid);
                else left = nullptr;
                if(ssize(div[0])) right = new Node(div[0], t, mid+1,
                    high);
                else right = nullptr;
            }
            else {
                left = right = nullptr;
            }
        }
    };

    Node *root;
    wavelet_tree(const vector<int> &b) {
        vector<int> idx;
        REP(i, ssize(b)) idx.eb(i);
        root = new Node(idx, b);
    };

    int cnt(Node *v, int l, int r) {
        if(v == nullptr) return 0;
        return int(upper_bound(all(v->v), r)-lower_bound(all(v->v),
            l));
    }

    int rank(int l, int r, int k, Node *v) {
        if(v->low == v->high) return v->low;
        if(cnt(v->left, l, r) >= k) return rank(l, r, k, v->left);
        else return rank(l, r, k-cnt(v->left, l, r), v->right);
    }

    int rank(int l, int r, int k) {
        return rank(l, r, k, root);
    }

    void clear(Node *v) {
```

```
        if(v->left != nullptr) clear(v->left);
        if(v->right != nullptr) clear(v->right);
        delete v;
    }
    ~wavelet_tree() {
        clear(root);
    }
};
```

treap.cpp
Opis: $O(\log n)$ czasowo. Treap z operacjami na (multi)secie. kth(t, k) zwraca k-tą najmniejszą liczbę.

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int val, rank;
        int size;
        Node(): left(nullptr), right(nullptr), val(0), rank(int(rng
            ())), size(0) {}

    };
    using pNode = Node*;

    Node pool[1'000'003];
    int curr = 0;

    int sz(pNode t) {
        return t == nullptr ? 0 : t->size;
    }

    void upd(pNode t) {
        t->size = sz(t->left) + sz(t->right) + 1;
    }

    pNode merge(pNode a, pNode b) {
        if(a == nullptr) return b;
        if(b == nullptr) return a;

        if(a->rank < b->rank) {
            a->right = merge(a->right, b);
            upd(a);
            return a;
        }
        else {
            b->left = merge(a, b->left);
            upd(b);
            return b;
        }
    }

    pair<pNode, pNode> split(pNode t, int x) {
        if(t == nullptr) return {nullptr, nullptr};

        if(t->val <= x) {
            auto [a, b] = split(t->right, x);
            t->right = a;
            upd(t);
            return {t, b};
        }
        else {
            auto [a, b] = split(t->left, x);
            t->left = b;
            upd(t);
            return {a, t};
        }
    }

    void insert(pNode &root, int x) {
```

```
pNode u = pool + (curr++);
u->val = x;
u->size = 1;
auto [a, b] = split(root, x);
root = merge(merge(a, u), b);
}

bool find(pNode t, int x) {
    if(t == nullptr) return false;
    if(x == t->val) return true;
    if(x < t->val) return find(t->left, x);
    else return find(t->right, x);
}

void erase(pNode root, int x) {
    auto [ab, c] = split(root, x);
    auto [a, b] = split(ab, x-1);
    root = merge(a, c);
}

int kth(pNode t, int k) {
    assert(k >= 0);
    if(sz(t->left) == k-1) return t->val;
    if(k > sz(t->left)) return kth(t->right, k-1-sz(t->left));
    else return kth(t->left, k);
}
}
```

implicit-treap.cpp

Opis: $O(\log n)$ czasowo. Implicit treap z podstawowymi operacjami, przykładowo odwracanie na przedziale (może być dowolna operacja). insert(root, k, x) wstawia element o wartości x w k -te miejsce tablicy. rev(root, l, r) odwraca przedział $[l; r]$ (indeksy od 0).

```
namespace Treap {
    mt19937 rng(420);
    struct Node {
        Node *left, *right;
        int rank;
        int size;
        bool rev;
        int val;
        Node(): left(nullptr), right(nullptr), rank(int(rng())),
            size(0), rev(false), val(0) {}
    };
    using pNode = Node*;

    Node pool[300'003];
    int curr = 0;

    int sz(pNode t) {
        return t == nullptr ? 0 : t->size;
    }

    void upd(pNode t) {
        t->size = sz(t->left) + sz(t->right) + 1;
    }

    void push_lazy(pNode t) {
        if(t->rev == false) return;
        if(t->left) t->left->rev = !t->left->rev;
        if(t->right) t->right->rev = !t->right->rev;
        swap(t->left, t->right);
        t->rev = false;
    }

    pNode merge(pNode a, pNode b) {
        if(a == nullptr) return b;
        if(b == nullptr) return a;
```

```
push_lazy(a);
push_lazy(b);
if(a->rank < b->rank) {
    a->right = merge(a->right, b);
    upd(a);
    return a;
}
else {
    b->left = merge(a, b->left);
    upd(b);
    return b;
}
}

pair<pNode, pNode> split(pNode t, int x) {
    if(t == nullptr) return {nullptr, nullptr};

    push_lazy(t);
    if(sz(t->left)+1 <= x) {
        auto [a, b] = split(t->right, x-(sz(t->left)+1));
        t->right = a;
        upd(t);
        return {t, b};
    }
    else {
        auto [a, b] = split(t->left, x);
        t->left = b;
        upd(t);
        return {a, t};
    }
}

pNode build(vector<int> a) {
    pNode root = nullptr;
    int n = ssize(a);
    for(int i = 0; i < n; ++i) {
        pNode u = pool + i;
        u->size = 1;
        u->val = a[i];
        root = merge(root, u);
    }
    curr += n;
    return root;
}

void insert(pNode &root, int k, int x) {
    auto [a, b] = split(root, k-1);
    Node *u = pool + (curr++);
    u->size = 1;
    u->val = x;
    root = merge(merge(a, u), b);
}

void reverse(pNode &root, int l, int r) {
    auto [ab, c] = split(root, r+1);
    auto [a, b] = split(ab, l);
    b->rev = !b->rev;
    root = merge(a, merge(b, c));
}

void print(pNode t) {
    if(t == nullptr) return;
    push_lazy(t);
    print(t->left);
    cout << t->val << ' ';
    print(t->right);
}
}
```

uf.cpp

Opis: $O(\log^* n)$ czasowo. Jeżeli $e[x] < 0$ to x jest korzeniem, a $-e[x]$ to rozmiar spójnej. Jeżeli $e[x] \geq 0$ to jest to ojciec x .

```
struct UF {
    vector<int> e;
    UF(int n):e(n, -1) {}

    int get(int a) {return e[a] < 0 ? a : e[a] = get(e[a]);}

    void unionn(int a, int b) {
        a = get(a); b = get(b);
        if(a == b) return;
        if(-e[a] > -e[b]) swap(a, b);
        e[b] += e[a];
        e[a] = b;
    }
};
```

dynamic-connectivity.cpp

Opis: $O(\log n)$ czasowo. Union find z roll backami – roll_back(x) cofa ostatnie x operacji unionn (union wierzchołków z tego samego zbioru się nie liczy).

```
struct dynamicUF {
    vector<int> e;
    dynamicUF(int n):e(n, -1) {}
    stack<array<int, 3>> s;

    int get(int a) {
        while(e[a] >= 0) a = e[a];
        return a;
    }

    int unionn(int a, int b) {
        a = get(a); b = get(b);
        if(a == b) return 0;
        if(-e[a] > -e[b]) swap(a, b);
        s.push({a, b, e[a]});
        e[b] += e[a];
        e[a] = b;
        return 1;
    }

    void roll_back(int m = 1) {
        while (m--) {
            auto [a, b, sz] = s.top();
            s.pop();
            e[b] -= sz;
            e[a] = sz;
        }
    }
};

fenwick-tree.cpp
Opis: (Binary Indexed Tree)  $O(\log N)$ , indeksowane od 0. Update w punkcie, zapytanie na przedziale
```

```
struct BIT {
    vector<int> tree;
    int N;
    BIT() {N = 0};
    BIT(int n) {
        N = n;
        tree.resize(N);
    }
    void resize(int n) {
        N = n;
        tree.resize(N);
    }
}
```

```
void update(int p, int x) {
    for (++p; p <= N; p += (p&-p)) tree[p-1] += x;
}
int sum(int r) {
    int res = 0;
    for (; r > 0; r -= (r&-r)) res += tree[r-1];
    return res;
}
int query(int l, int r) {return sum(r+1)-sum(l);}
int lower_bound(int sum) {
    int pos = 0;
    for (int pw = 1<<25; pw; pw<<=1) {
        int npos = pos+pw;
        if (npos <= N && tree[npos-1] < sum) {
            pos = npos; sum -= tree[pos-1];
        }
    }
    return pos;
};
```

Optymalizacje (3)

3.1 FastOut

Do każdego kodu dopisany jest przykładowy czas (wypisanie 10⁷ liczb). Plik fastout.cpp wypisuje dodatkowe znaki białe, uważa że na rozmiar bufora. Flushować bufor jak trzeba, ale podzielić wczytywanie (nie dodawać ifa w środku żadnej funkcji, bo będzie wolno).

fastout.cpp

Opis: Trzeba wywołać flush_out() żeby wszystko wypisać.
czas: ok. 0,067s na time, 195ms na oiejq.

```
const int BUF_SIZE = 1'000'100 * 22;
char buf[BUF_SIZE];
char *buf_ptr = buf;

constexpr unsigned long
build_long(char c1, char c2, char c3, char c4, char c5, char c6
, char c7, char c8) {
    unsigned long n = c8;
    n <<= 8;
    n |= c7;
    n <<= 8;
    n |= c6;
    n <<= 8;
    n |= c5;
    n <<= 8;
    n |= c4;
    n <<= 8;
    n |= c3;
    n <<= 8;
    n |= c2;
    n <<= 8;
    n |= c1;
    return n;
}

template <bool LeadingZeroes> struct output {

    constexpr output() {
        char lead = LeadingZeroes ? '0' : ' ';

        for(unsigned long c1 = 1; c1 <= 9; ++c1)
            for(unsigned long c2 = 0; c2 <= 9; ++c2)
```

```
for(unsigned long c3 = 0; c3 <= 9; ++c3)
    for(unsigned long c4 = 0; c4 <= 9; ++c4)
        for(unsigned long c5 = 0; c5 <= 9; ++c5)
            arr[10'000*c1+1'000*c2+100*c3+10*c4+c5] =
                build_long('0'+c1, '0'+c2, '0'+c3, '0'+c4, '0'+c5, '0', '0');

for(unsigned long c2 = 1; c2 <= 9; ++c2)
    for(unsigned long c3 = 0; c3 <= 9; ++c3)
        for(unsigned long c4 = 0; c4 <= 9; ++c4)
            for(unsigned long c5 = 0; c5 <= 9; ++c5)
                arr[1'000*c2+100*c3+10*c4+c5] = build_long(lead, '0'+c2, '0'+c3, '0'+c4, '0'+c5, '0', '0');

for(unsigned long c3 = 1; c3 <= 9; ++c3)
    for(unsigned long c4 = 0; c4 <= 9; ++c4)
        for(unsigned long c5 = 0; c5 <= 9; ++c5)
            arr[100*c3+10*c4+c5] = build_long(lead, lead, '0'+c3, '0'+c4, '0'+c5, '0', '0');

for(unsigned long c4 = 1; c4 <= 9; ++c4)
    for(unsigned long c5 = 0; c5 <= 9; ++c5)
        arr[10*c4+c5] = build_long(lead, lead, lead, '0'+c4, '0'+c5, '0', '0');

for(unsigned long c5 = 0; c5 <= 9; ++c5)
    arr[c5] = build_long(lead, lead, lead, lead, '0'+c5, '0', '0');
}

array<unsigned long, 100'000> arr{};
};

constexpr output<true> out_lz;
constexpr output<false> out_ls;

void write_long(unsigned long l, unsigned long n) {
    reinterpret_cast<unsigned long*> (buf_ptr) = l;
    buf_ptr += n;
}

pair<unsigned long, unsigned long> podziel(unsigned long a, unsigned long b) {
    return {a/b, a%b};
}

void write_int(unsigned long x) {
    if(x >= 1'000'000'000'000'000) {
        auto [abc, d] = podziel(x, 100'000);
        auto [ab, c] = podziel(abc, 100'000);
        auto [a, b] = podziel(ab, 100'000);
        write_long(out_ls.arr[a], 5);
        write_long(out_lz.arr[b], 5);
        write_long(out_lz.arr[c], 5);
        write_long(out_lz.arr[d], 6);
    } else if(x >= 10'000'000'000) {
        auto [ab, c] = podziel(x, 100'000);
        auto [a, b] = podziel(ab, 100'000);
        write_long(out_ls.arr[a], 5);
        write_long(out_lz.arr[b], 5);
        write_long(out_lz.arr[c], 6);
    }
    else if(x >= 100'000) {
        auto [a, b] = podziel(x, 100'000);
        write_long(out_ls.arr[a], 5);
        write_long(out_lz.arr[b], 6);
    }
    else {
        write_long(out_ls.arr[x], 6);
    }
}
```

```
}

void flush_out() {
    write(STDOUT_FILENO, buf, buf_ptr-buf);
    buf_ptr = buf;
}
```

short-out.cpp

Opis: Trzeba wywołać flush_out() żeby wszystko wypisać.
czas: ok. 0,47s na time, 1955ms na oiejq.

```
char buf[200 * 1'024 * 1'024];
char *buf_ptr = buf;

void write_char(char c) {
    *buf_ptr++ = c;
}

pair<unsigned long, unsigned long> podziel(unsigned long a, unsigned long b) {
    return {a/b, a%b};
}

void write_int_petla(unsigned long x) {
    do {
        auto [c, d] = podziel(x, 10);
        write_char('0'+d);
        x = c;
    } while (x > 0) ;
}

void write_int(unsigned long x) {
    char *old_buf_ptr = buf_ptr;
    write_int_petla(x);
    reverse(old_buf_ptr, buf_ptr);
    write_char('\n');
}

void flush() {
    write(STDOUT_FILENO, buf, buf_ptr-buf);
    buf_ptr = buf;
}
```

no-buf-out.cpp

Opis: czas: ok. 0,67s na time, 2816ms na oiejq.

```
void write_int(unsigned long x) {
    char t[19];
    int i = 0;
    do {
        int d = x%10;
        t[i++] = '0'+d;
        x /= 10;
    } while (x > 0) ;

    while (--i >= 0) {
        putchar_unlocked(t[i]);
    }
    putchar_unlocked('\n');
}
```

3.2 Fastin

Do każdego kodu dopisany jest przykładowy czas (wczytanie 2 · 10⁷ liczb).

fastin.cpp

Opis: Trzeba wywołać read_buf() na początku maina.
czas: 0,15s na time, 772ms na oiejq.

```
const int BUF_IN_SIZE = 200 * 1'024 * 1'024;
char buf_in[BUF_IN_SIZE];
int pos;

bool is_digit(char c) {
    return c >= '0';
}

bool is_space(char c) {
    return c == ' ';
}

int fastin(){
    int x = 0;
    char c;
    while (is_space(c = buf[pos++])) ;
    pos--;
    while (is_digit(c = buf_in[pos++])) {
        x = x * 10 + c - '0';
    }
    return x;
}

void read_buf() {
    read(STDIN_FILENO, buf_in, BUF_IN_SIZE);
    pos = 0;
}
```

no-buf-in.cpp
Opis: czas: 0,38s na time, 1052ms na oiejq.

```
bool is_digit(char c) {
    return c >= '0';
}

int fastin(){
    int x = 0;
    char c;
    while (is_digit(c = getchar_unlocked())) {
        x = x * 10 + c - '0';
    }
    return x;
}
```